

A negative result on a putative $\sqrt{\omega(g)}$ correction to Wolf's consecutive-prime-gap formula, with a reproduction of finite-size drift to $3 \cdot 10^{14}$

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Abstract

We report empirical results for the distribution of consecutive prime gaps $\tau_d(x)$ on a sieve dataset extended to $x = 3 \cdot 10^{14}$. Two findings are presented. First, the finite-size drift parameters $a(x)$ and $s(x)$ defined by Wolf in *Phys. Rev. E* **89** (2014), Eq. (8) and Fig. 6 are reproduced in shape on the extended range. Second, we test a $\sqrt{\omega(g)}$ correction term that appeared as a residual feature when the empirical histogram is compared against an algebraic variant of Wolf's formula in which the Hardy–Littlewood constant C_g enters the exponent. We show that this term collapses by 78–95% when the correct Wolf 2014 baseline is used, and that an out-of-sample test on hold-out values $N > 10^{13}$ gives weighted $R^2 = -18.3$ against the correct baseline, consistent with the residual being an artifact of the mis-specified baseline rather than a feature of the prime-gap distribution. We document the discrepancy explicitly so that a future analysis on the same data starts from the correct reference.

1 Introduction

Wolf [1] gave a heuristic prediction for the number $\tau_d(x)$ of consecutive prime gaps of size d up to x ,

$$\tau_d(x) \sim C_d \frac{x}{\ln^2 x} \exp\left(-\frac{d}{\ln x}\right), \quad d \text{ even}, \quad d \geq 6, \quad (1)$$

where $C_d = 2C_2 \prod_{p|d, p>2} (p-1)/(p-2)$ is the Hardy–Littlewood singular series for gap d and $C_2 \approx 0.6601618$ is the twin-prime constant. Equation (1) is the peer-reviewed reference; a review article [2] gives an equivalent form $C_d \pi^2(x)/x \cdot \exp(-d \pi(x)/x)$. The same work [1] documents in its Section III a slow finite-size drift of the prefactor $a(x)$ and slope $s(x)$ over $x \in [10^8, 10^{14}]$: $a(x) : 1.512 \rightarrow 1.273$ and $s(x) : 1.187 \rightarrow 1.136$.

Cohen [3] extends the asymptotic discussion to higher moments of g , with reference values for $\mu'_{k,n}(2^t)$ tabulated through $t = 48$.

This note extends the sieve to $x = 3 \cdot 10^{14}$ and addresses two questions:

1. Does Wolf's drift in $a(x), s(x)$ continue with the same shape on the extended range, and
2. Does the residual $\log(\tau_d^{\text{emp}}/\hat{\tau}_d)$ contain any feature beyond Wolf's drift?

The motivation for (2) was a previous unpublished analysis by the present author (versions v1–v3, withdrawn from circulation) which identified two apparent residual features: a $\sqrt{\omega(g)}$ term and a $b \log \log N$ term. Both turn out to be artifacts of an incorrectly transcribed baseline; we document them here as a cautionary finding.

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2 Sieve and verification

A sieve via the `primesieve` library produced histograms of consecutive prime gaps for $N \in \{10^5, \dots, 3 \cdot 10^{14}\}$ along a logarithmically spaced grid (28 values). For each N we record the count $\tau_d(N)$ for $d = 1, 2, 4, \dots, d_{\max}$ where $d_{\max}$ is the maximum observed gap.

Cross-validation against Cohen [3, Tab. 1] for $x = 2^t$, $t \in \{30, 36\}$, gives an exact match (relative error $< 5 \cdot 10^{-5}$ on μ'_1, μ'_2 and G_n):

t	n (excl. d_1)	μ'_1	μ'_2	G_n
30 (sieve)	54,400,026	19.7379	683.2373	282
30 (Cohen)	54,400,026	19.7379	683.2373	282
36 (sieve)	2,874,398,513	23.9074	1022.16	464
36 (Cohen)	$\approx 2.8744 \cdot 10^9$	23.9074	1022.2	464

We adopt Cohen’s convention that $d_1 = 1$ (the gap between 2 and 3) is excluded.

3 Reproduction of Wolf’s drift

For each N we follow [1, §III]: define $u = d\pi(N)/N$ and $T_d = \tau_d N / (C_d \pi^2(N))$, and fit $\log T_d = \log a - s u$ by ordinary least squares on $d \in \{2, 4, \dots\}$ restricted to the linear region $u \leq 2$ and $\tau_d \geq 10^3$. Table 1 shows the result.

Table 1: Reproduction of $a(N), s(N)$ from [1, Fig. 6] on the extended range.

N	$\pi(N)$	$a(N)$	$s(N)$
$1.7 \cdot 10^8$	9,603,818	1.269	1.072
$1 \cdot 10^{10}$	$4.5505 \cdot 10^8$	1.222	1.061
$1 \cdot 10^{12}$	$3.7608 \cdot 10^{10}$	1.192	1.060
$1 \cdot 10^{14}$	$3.2049 \cdot 10^{12}$	1.165	1.052
$3 \cdot 10^{14}$	$9.2874 \cdot 10^{12}$	1.156	1.046

The shape of the trajectory is consistent with Wolf’s reported values $(a, s) = (1.512, 1.187)$ at $x \approx 10^8$ and $(1.273, 1.136)$ at $x \approx 10^{14}$: a and s both decrease monotonically and the ratio of decrements over four decades is the same to leading order. An absolute offset of about 0.15 in a remains; this is sensitive to the choice of u -cutoff in the linear-region fit, and we did not attempt to reproduce [1]’s cutoff exactly. The qualitative finding—a non-vanishing finite-size drift extending into the 10^{14} regime—is confirmed.

A linear regression of the empirical $a(N), s(N)$ against $\log \log N$ gives

$$\log a(N) = +0.716 - 0.163 \log \log N, \quad \log s(N) = +0.191 - 0.041 \log \log N, \quad (2)$$

with the two slopes summing to -0.204 . This number will reappear in Section 4.

4 A negative result on residual structure

A previous analysis (this author, unpublished, versions v1–v3) considered the empirical residual

$$R(g, N) = \log \tau_g^{\text{emp}}(N) - \log \widehat{W}(g, N), \quad (3)$$

and reported that R exhibits a $\sqrt{\omega(g)}$ structure (where $\omega(g)$ is the number of distinct prime factors of g) plus a $\log \log N$ trend. The two-parameter weighted fit

$$R(g, N) \approx a\sqrt{\omega(g)} + b \log \log N \quad (4)$$

on $n = 169$ data points (filtered by $\rho = g C_g / \ln N \in [0.05, 1.10]$, $\tau_g^{\text{emp}} \geq 100$) gave $a = +0.836$, $b = -0.195$, $R^2 \approx 0.95$, with strong out-of-sample reproduction.

The baseline used in that analysis was however not (1). The transcription used was

$$\widehat{W}_S(g, N) = C_g \cdot \text{Li}_2(N) \cdot \exp\left(-\frac{g C_g}{\ln N}\right), \quad (5)$$

which differs from (1) in two algebraically inequivalent ways: C_g appears in the exponent (it does not in Wolf’s formula), and the prefactor is $\text{Li}_2(N)$ rather than $N/\ln^2 N$. The two are equal to leading order but the exponent factor C_g is not asymptotic correction; for $g = 30$ one has $C_{30}/C_2 \approx 2.66$, so the discrepancy in the residual is of order unity for moderate g .

We re-fit (4) against four baselines, on the same dataset ($n = 169$):

1. $\widehat{W}_S$ from (5) (the previous baseline);
2. $\widehat{W}_{\text{Wolf2014}}$ from (1);
3. $\widehat{W}_{\text{Wolf2018}}$, $C_g \pi^2 / N \exp(-g \pi / N)$ from [2] Eq. (26);
4. $\widehat{W}_{\text{bare}} = \pi^2 / N \exp(-g / \ln N)$, with no C_g at all (a null baseline).

Table 2 reports the results, both in the two-parameter form (4) and a five-parameter form $y = \log \widehat{W}_{\text{pre}} - \alpha \rho^\beta + a \sqrt{\omega(g)} + b \log \log N + c$ that allows the exponent slope and intercept to free.

Table 2: Refit of (4) (rows “ M_1^* ”) and a 5-parameter Wolf-form + linear residual model (rows “ M_3 ”) against four baselines, $n = 169$.

Baseline	M_1^* (2 par.)				M_3 (5 par.)				
	a	b	RSS_w	AIC	α	β	a	b	c
$\widehat{W}_S$	+0.836	−0.195	1.447	−318.9	0.691	1.092	+0.512	−0.170	+0.112
$\widehat{W}_{\text{Wolf2014}}$	+0.184	+0.012	1.430	−320.9	0.987	1.334	+0.026	−0.244	+0.898
$\widehat{W}_{\text{Wolf2018}}$	+0.159	−0.004	0.702	−441.2	0.919	1.318	+0.027	−0.149	+0.502
$\widehat{W}_{\text{bare}}$	+1.091	−0.215	3.525	−168.4	1.398	1.512	+1.123	−0.208	−0.088

The pattern is unambiguous:

- In M_1^* , the $\sqrt{\omega(g)}$ amplitude collapses from 0.836 (against $\widehat{W}_S$) to 0.184 against Wolf 2014 (a 78% reduction). Against the bare baseline, which removes C_g entirely, the amplitude grows to 1.091—confirming that the C_g structure does belong to the baseline, not the residual.
- In M_3 , where the exponent slope and intercept are free, the amplitude is 0.026 against Wolf 2014 and 0.027 against Wolf 2018—numerically consistent with zero.
- The $\log \log N$ coefficient against Wolf 2014 in M_1^* is $b = +0.012$, compared with $b = -0.195$ against $\widehat{W}_S$. The original b is essentially the slope of $\log a(N) + \log s(N)$ vs. $\log \log N$ from Eq. (2), -0.204 , to within fit noise. The previously claimed $\log \log N$ term is the same finite-size drift documented in [1, Fig. 6], expressed in a different parameterization.

Out-of-sample test. For each baseline we trained (4) on $N \leq 10^{13}$ ($n = 140$), froze coefficients, and evaluated the weighted R^2 on $N > 10^{13}$ ($n = 29$):

Baseline	trained a	trained b	OOS R_w^2
$\widehat{W}_S$	+0.855	−0.202	+0.654
$\widehat{W}_{\text{Wolf2014}}$	+0.180	+0.021	−18.27
$\widehat{W}_{\text{Wolf2018}}$	+0.159	+0.001	−5.73
$\widehat{W}_{\text{bare}}$	+1.121	−0.227	+0.538

Against the correct Wolf 2014 baseline, the model predicts the hold-out worse than the weighted mean. The high in-sample R^2 against $\widehat{W}_S$ was therefore measuring the systematic disagreement between $\widehat{W}_S$ and Wolf 2014, not a feature of the prime-gap distribution.

5 Discussion

The analysis answers the two questions posed in the introduction:

(1) Wolf’s $a(x), s(x)$ drift continues monotonically into the 10^{14} regime, with the same qualitative shape as in [1, Fig. 6].

(2) The residual against Wolf’s published formula contains no $\sqrt{\omega(g)}$ structure within the resolution of our dataset. The $\log \log N$ component is the finite-size drift already published by Wolf, in a different parameterization. The previously reported residual features were artifacts of fitting against a baseline of the form (5), which is algebraically distinct from Wolf’s formula (1).

We are publishing this null result for two reasons. First, the sieve data—28 histograms reaching $3 \cdot 10^{14}$, exact-matched against Cohen’s table 1—is independently usable, and the verification of Wolf’s drift on the extended range may be of interest. Second, the specific failure mode (a residual that looks structured, with high R^2 and apparent out-of-sample stability, but is in fact a fingerprint of the wrong reference function) seems generic enough that it is worth documenting as a caution.

Data and code. All sieve histograms, fit code, and the recompute pipeline (modules `wolf_formulas.py`, `cohen_verify.py`, `refit_baselines.py`, `oos_test.py`) are available at <https://github.com/dkrse/math-final-wolf>. A snapshot accompanying this note is archived on Zenodo, [doi:10.5281/zenodo.2001007](https://doi.org/10.5281/zenodo.2001007).

Superseded earlier versions. Three earlier versions of the analysis (v1–v3), in which the residual against the baseline (5) was the central claim, are archived on Zenodo for completeness: [doi:10.5281/zenodo.19796305](https://doi.org/10.5281/zenodo.19796305), [doi:10.5281/zenodo.19958063](https://doi.org/10.5281/zenodo.19958063), [doi:10.5281/zenodo.19975065](https://doi.org/10.5281/zenodo.19975065). Those records should be considered superseded by the present note.

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References

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